

CCNA 200-301 Notes

Chapter 03 - Network Fundamentals (Part 3)

Topics Covered

- ARP Process
- MAC Address Learning
- Switch Forwarding
- Collision Domain vs Broadcast Domain
- Hub vs Switch vs Router

Key Notes

- When a host knows the Destination IP but not the Destination MAC, it sends an ARP Request (Broadcast).
- Only the target device replies with an ARP Reply (Unicast).
- The sender stores the IP ↔ MAC mapping in the ARP Cache.
- A switch learns only the Source MAC Address and stores it in the MAC Address Table (CAM Table).
- If the Destination MAC is unknown, the switch floods the frame to all ports except the incoming port.
- Once the MAC is learned, future frames are forwarded only to the correct port.

Collision Domain vs Broadcast Domain

Device	Collision Domain	Broadcast Domain
Hub	1	1
Switch	One per Port	1 (Default VLAN)
Router	One per Interface	One per Interface

Exam Tips ☐

- ARP Request = Broadcast
- ARP Reply = Unicast
- Switch learns Source MAC only

- Unknown Destination MAC = Flood
- Router does NOT forward Broadcast traffic

Useful Commands

- show mac address-table
- show arp